Steps of the Machine Learning

- Data Collection
- EDA
- Transformation
- Model Training
- Model Deployment

EDA(Pre-processing)

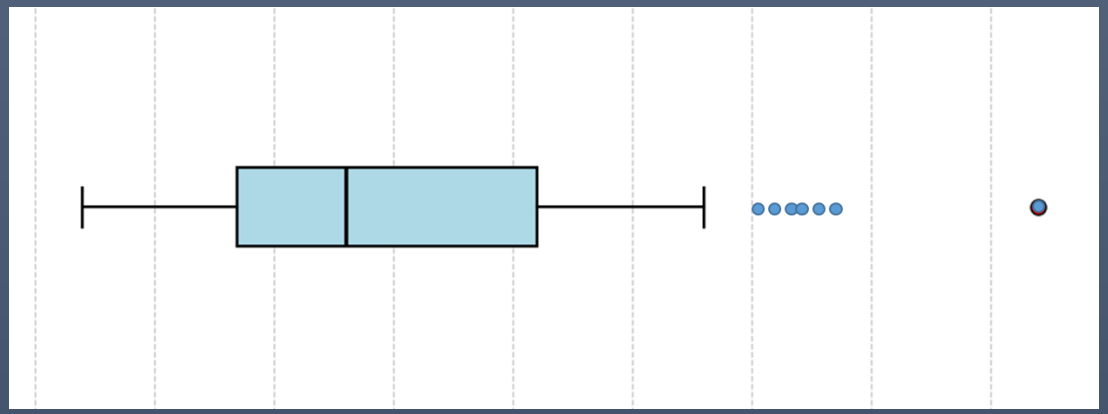
1) Data Cleaning:
2) Data Analysis

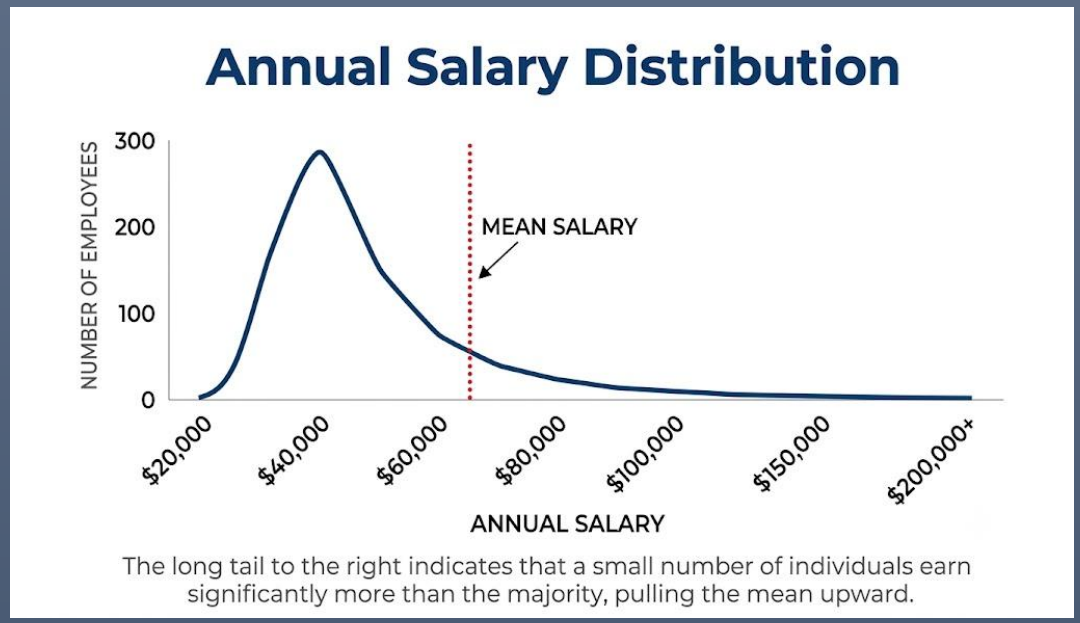
Data Cleaning

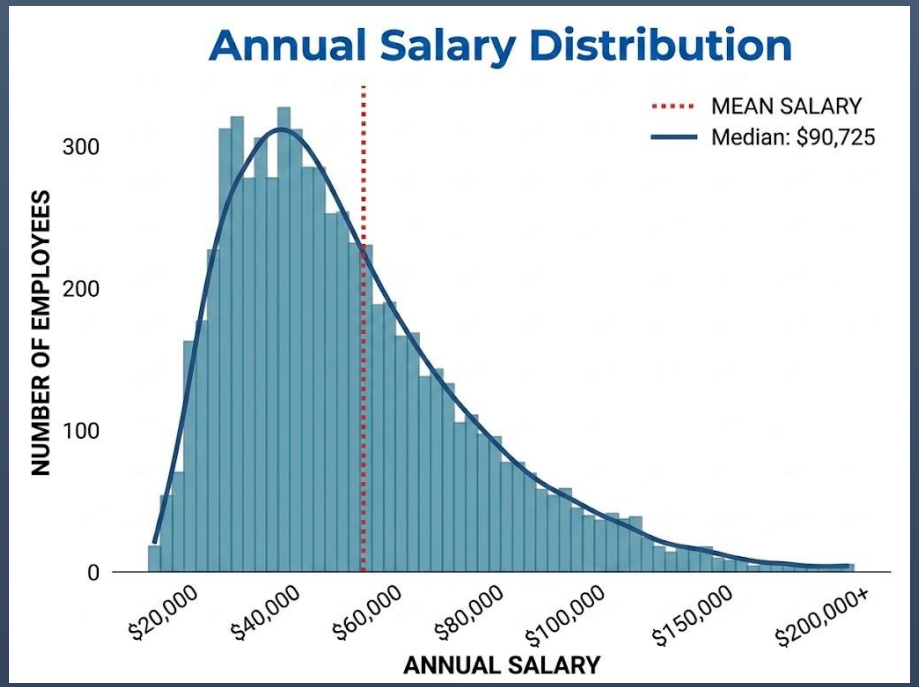
- Handle Missing Value:
    1) Random value
    2) Forward filling/backward filling
    3) **Statistical approach (mean. mode, median)**
    4) **Drop Missing Value**
    5) Create “your own” ML model to predict missing value(While using Advance Level)
- Handel Duplicate Values(Drop the duplicate)
- Handle uneven Values(Spelling Mistakes, old Name and New name)
- Handling Date Column
- Changing Data type of the columns based on requirement
- Increasing or Decreasing the columns based on existing columns
- **Outlier Handling**

Outlier Handling

An **outlier** is a data point that is significantly different from the remaining data points in a dataset. Think of it as the "odd one out"—it lies an abnormal distance from other values in a random sample.







Numerical method

- **IQR Method (Interquartile Range)**

Visual Inspection:

- **Box plots** (where outliers typically appear as isolated dots outside the "whiskers").
- **Scatter plots** (points that sit far away from the main cluster of data).

[95, 100, 125, 130, 135, 140, 145, 150, 170, 175, 180, 185, 200, 300, 305, 310]

$$Postion\ of\ Q_1 = \frac{n+1}{4}$$

$$Postion\ of\ Q_1 = \frac{16+1}{4}$$

$$Postion\ of\ Q_1 = \frac{17}{4}$$

$$Postion\ of\ Q_1 = 4.25$$

$$Q_1 = 130. + (130 - 135) * 0.25$$

$$Q_1 = 130 + (5) * 0.25$$

$$Q_1 = 130 + 1.25$$

$$Q_1 = 131$$

[95, 100, 125, 130, 135, 140, 145, 150, 170, 175, 180, 185, 200, 300, 305, 310]

$$Postion\ of\ Q_3 = \frac{3(n+1)}{4}$$

$$Postion\ of\ Q_3 = \frac{3(16+1)}{4}$$

$$Postion\ of\ Q_3 = \frac{3(17)}{4}$$

$$Postion\ of\ Q_3 = \frac{51}{4}$$

$$Postion\ of\ Q_3 = 12.75$$

$$Q_3 = 185 + (200 - 185) * 0.75$$

$$Q_3 = 185 + (15) * 0.75$$

$$Q_3 = 185 + 11.25$$

$$Q_3 = 196.25$$

[95, 100, 125, 130, 135, 140, 145, 150, 170, 175, 180, 185, 200, 300, 305, 310]

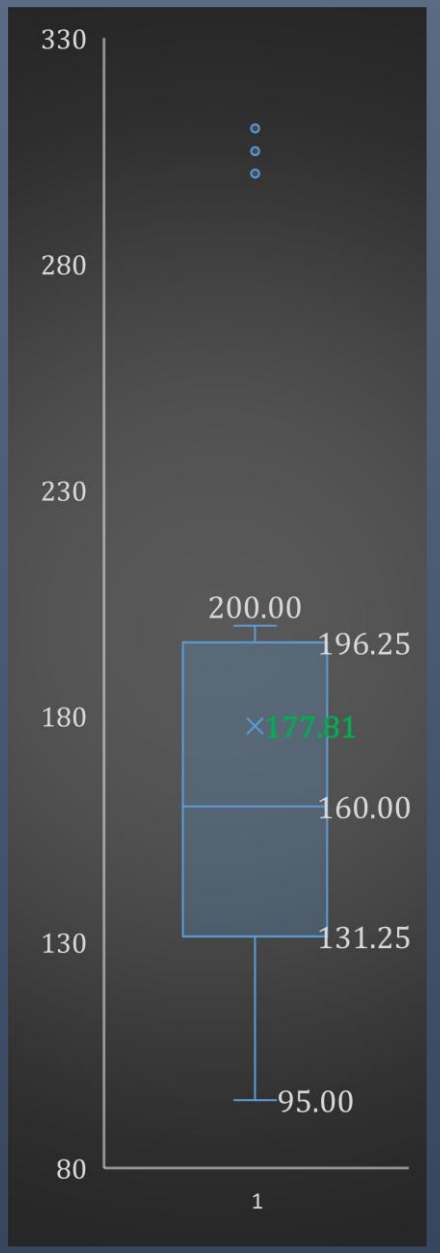
$iqr = Q_3 - Q_1$

$iqr = 196.25 - 131.25$

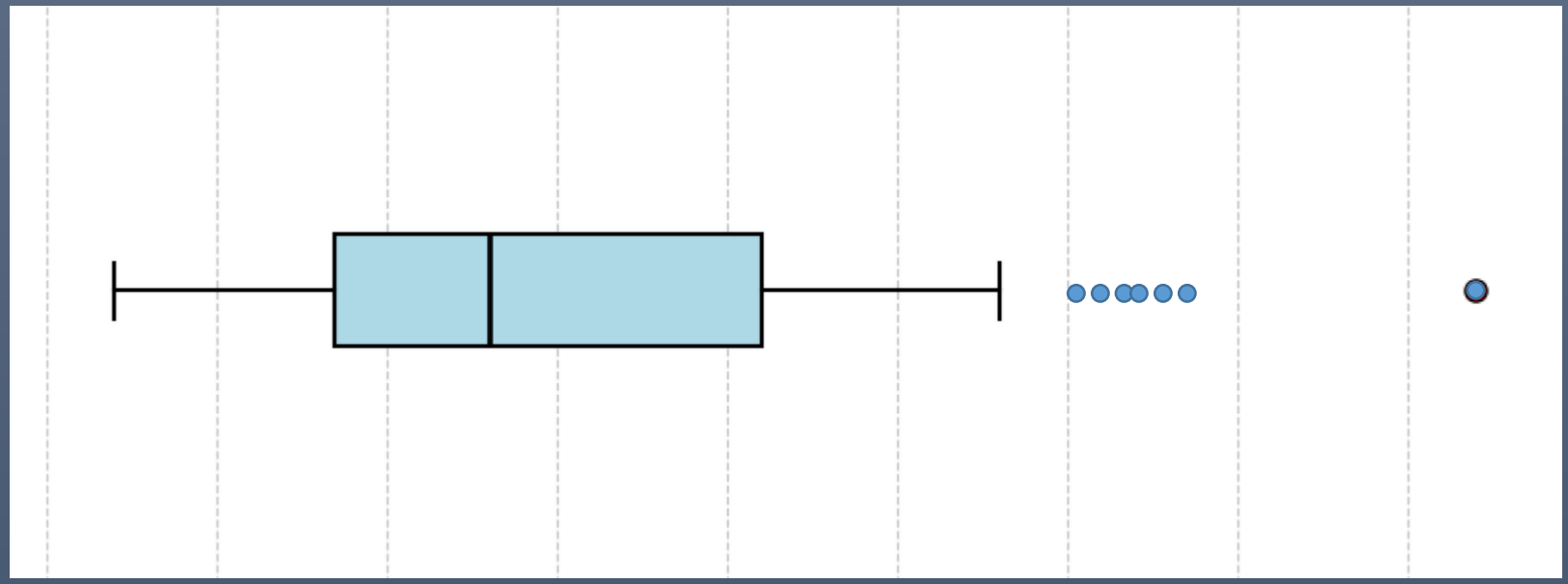
$iqr = 65$

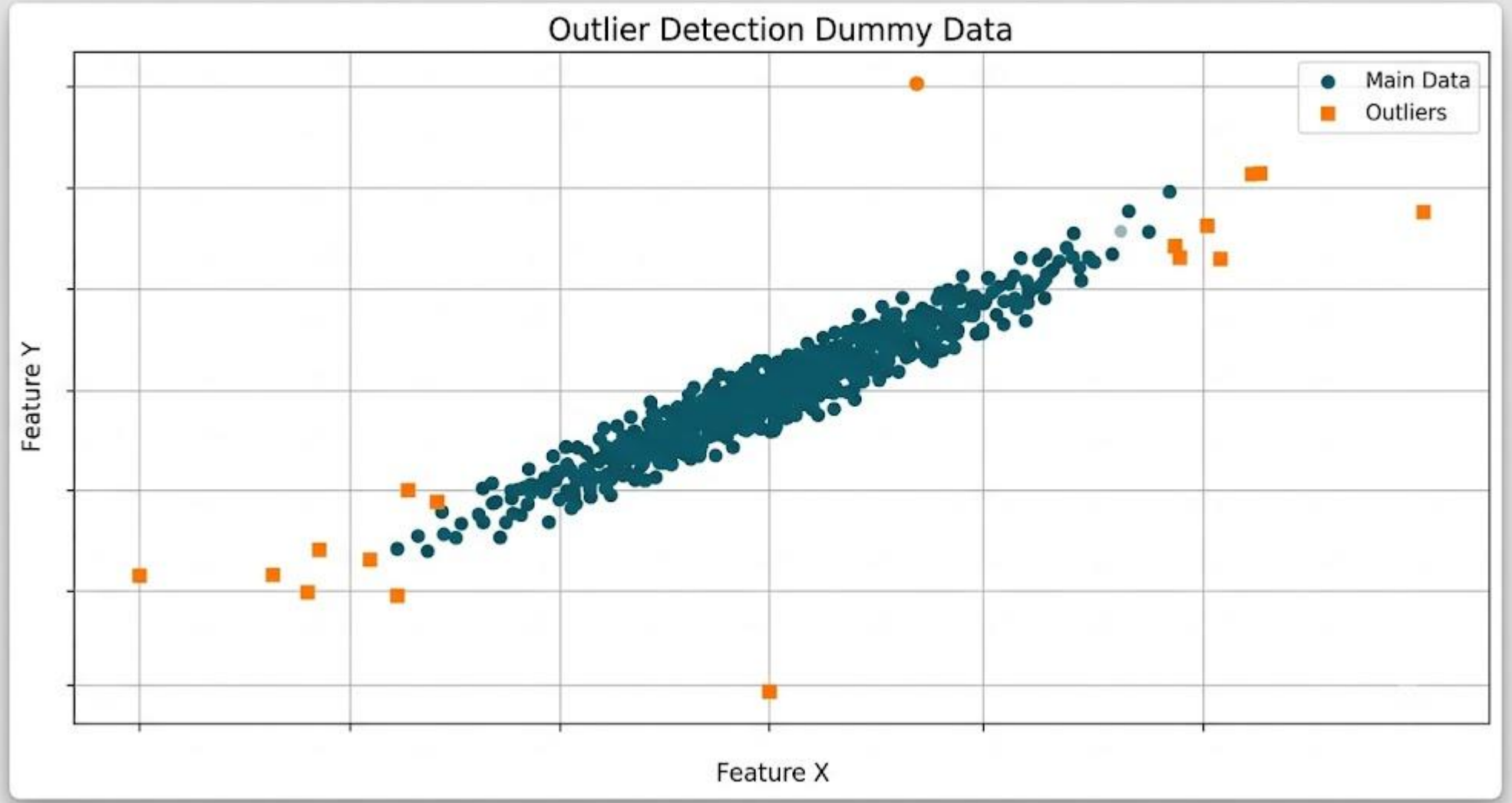
Lower Fence $= Q_1 - (1.5 * \text{IQR})$
$= 131.25 - (1.5 * 65)$
$= 131.25 - 97.5$
$=$ **33.75**

Upper Fence $= Q_3 + (1.5 * \text{IQR})$
$= 196.25 + (1.5 * 65)$
$= 196.25 + 97.5$
$=$ **293.75**



Box Plot




Outlier Detection Dummy Data
Main Data
Outliers
Feature Y
Feature X

Analysis

Stats or Numerical Method

- Univariate Analysis
- Bivariate analysis

Numerical Method

1) Univariate Analysis
    1) Measures of Central Tendency
    2) Measures of Dispersion
    3) Measure of Location
    4) Measures of Distribution Shape
    5) Frequency Distribution
2) Bivariate analysis (Tow variable)
    1) Measures of Association
        - Pearson correlation,
        - spearman rank correlation

Transformation

1) **Encoding**
2) **Scaling**

Advance Pre-processing

1) Handel Imbalance Data
2) Feature Selection
3) Dimension Reduction

1) Encoding

Encoding is the process of translating Categorical data into Numerical so that a ML algorithm can easily store, transmit, or understand it and perform some calculation on it.

1) **One hot Encoding**
2) **Ordinal Encoding**

One hot Encoding

Use one-hot encoding when:
The feature is **categorical and nominal** (e.g., Color, City, Gender).
The number of unique categories is **small or moderate**.

Student	Color
A	Red
B	Blue
C	Green
D	Red

```
from sklearn.preprocessing import OneHotEncoder
```

Student	Color_Red	Color_Blue	Color_Green
A	1	0	0
B	0	1	0
C	0	0	1
D	1	0	0

Ordinal Encoding

Use **Ordinal Encoding** when the categories have a **meaningful order**, such as:

Student	Size
A	Small
B	Medium
C	Large
D	Medium

Education	Encoded Value
High School	1
Bachelor's	2
Master's	3
PhD	4

Student	Size
A	1
B	2
C	3
D	2

```
from sklearn.preprocessing import OrdinalEncoder
```

Scaling

Scaling is the process of changing the range of numerical values so that features are on a similar scale.
The two most common scaling techniques are
Normalization and **Standardization**.

Normalization (Min Max Scaler)

Normalization is the process of organizing the data in the database. · Normalization is used to minimize the redundancy from a relation or set of relations.

It restricts the data from 0 to 1

Normalize the below data

X =[1,2,3,4,5,6,7,8,9,10]

$$y = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Values	Y
1-1/10-1	0.000
2-1/10-1	0.111
3-1/10-1	0.222
4-1/10-1	0.333
5-1/10-1	0.444
6-1/10-1	0.556
7-1/10-1	0.667
8-1/10-1	0.778
9-1/10-1	0.889
10-1/10-1	1.000

```
from sklearn.preprocessing import MinMaxScaler
```

Standardization (Standard Normal Distribution:)

Converting a **normal distribution** in to **standard normal distribution** is called Standardization. The standard normal distribution is also called the z-distribution; **any normal distribution can be standardized by converting its values into z-scores formula.**

Where mean become zero and standard deviation becomes one

Calculate the Standard Deviation:

X [1,2,3,4,5]

$$\text{Z Score} = \frac{x_i - \mu}{\sigma}$$

$\mu = \text{Mean}$

σ= Standard Deviation

1	$\frac{1-3}{1.41} =$	-1.414
2	$\frac{2-3}{1.41} =$	-0.707
3	$\frac{3-3}{1.41} =$	0
4	$\frac{4-3}{1.41} =$	0.707
5	$\frac{5-3}{1.41} =$	1.414

```
from sklearn.preprocessing import StandardScaler
```

$$y = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$\text{Z Score} = \frac{x_i - \mu}{\sigma}$$

Normalization	Standardization
The minimum and maximum values of features are used for scaling	Mean and the standard deviation is used for scaling.
It is used when features are of different scales.	It is used when we want to ensure zero mean and unit standard deviation.
Scales values between [0, 1] or [-1, 1].	It is not bounded to a certain range.
It is really affected by outliers.	It is much less affected by outliers.
It is useful when we don't know about the distribution	It is useful when the feature distribution is Normal or Gaussian.

Steps of the Machine Learning

- Data Collection
- EDA(Pre-processing)
    1) Data Cleaning
    2) Data Analysis
    3) Transformation
        - Encoding
        - Scaling
- Model Building
    1) Model Training
    2) Deployment

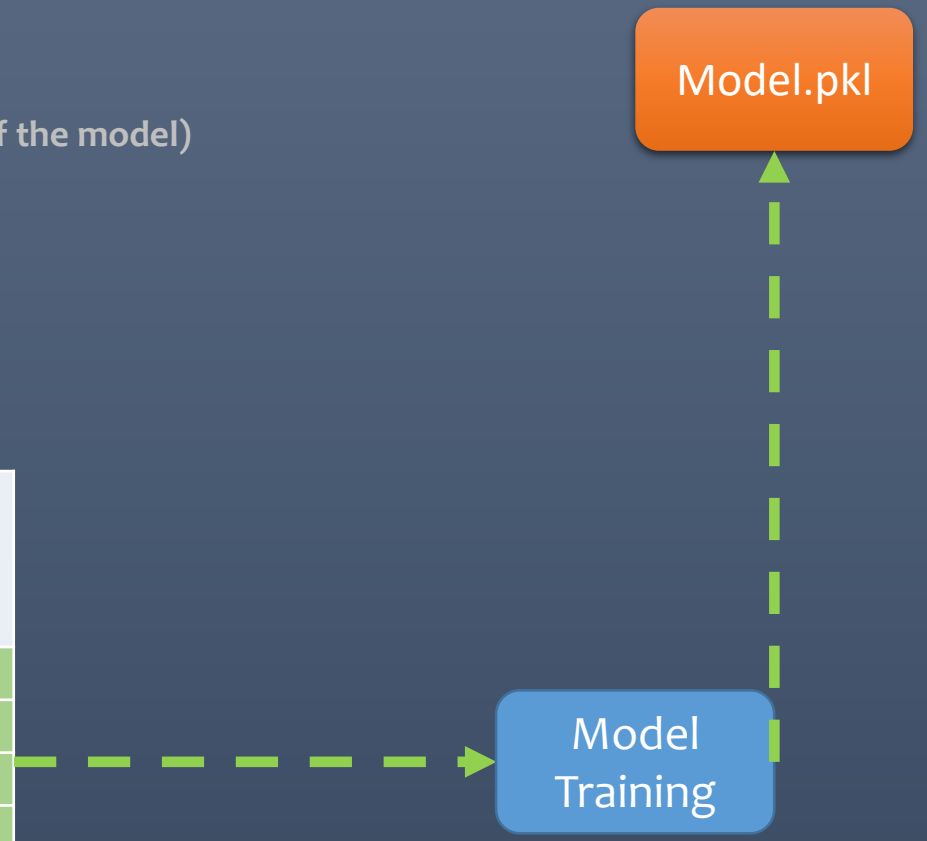
Model Building

- Training
- Deployment

Training

- Choose model based on problem based on problem
  - classification
  - regression
  - clustering
- Hyper parameter tuning
- Evaluation of the Model Use different test to know the performance of the model)
- Then decide the best model based on training and evaluation
- Save model as pkl file

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0



1. Choose a Model Based on the Problem

Select the machine learning algorithm according to the type of problem:

- **Classification:** Predicts categories or classes (e.g., Spam/Not Spam, Disease/No Disease).
- **Regression:** Predicts continuous values (e.g., House Price, Salary).
- **Clustering:** Groups similar data points without labels (e.g., Customer Segmentation).

2. Hyperparameter Tuning

- Adjust the model's hyperparameters to improve its performance.
- Hyperparameter tuning helps the model achieve better accuracy and generalization.
- Common methods include **Grid Search** and **Random Search**.

3. Evaluate the Model

- Test the trained model using different evaluation metrics to measure its performance.
- **Classification:** Accuracy, Precision, Recall, F1-Score.
- **Regression:** MAE, MSE, RMSE, R^2 Score.
- **Clustering:** Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Score.

4. Select the Best Model

- Compare the performance of different models.
- Choose the model that performs best based on the evaluation results and generalizes well to new data.

5. Save the Model as a .pkl File

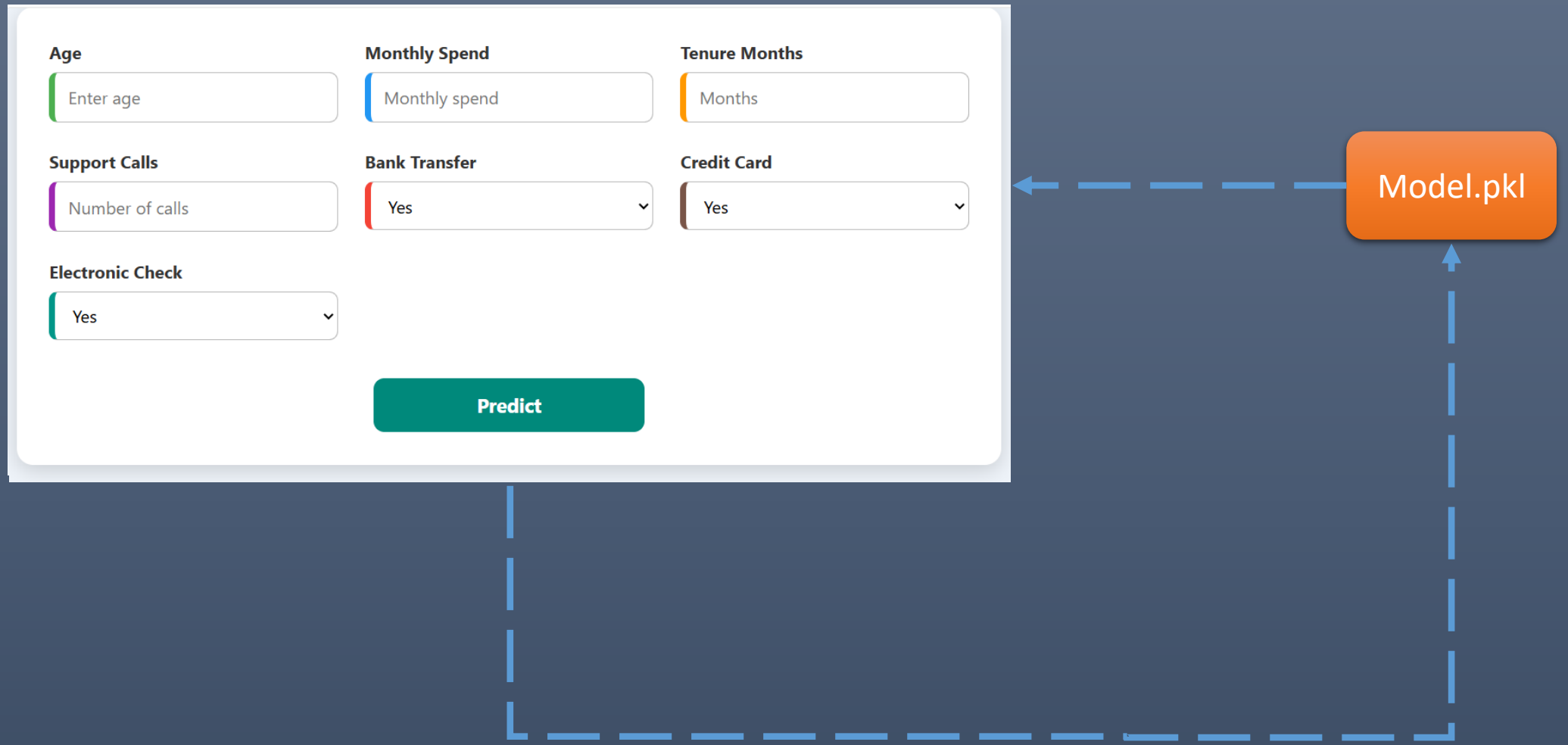
- Save the final trained model in a **.pkl (Pickle)** file.
- The saved model can be loaded later to make predictions without retraining.

Prediction Phase

Create an API for use so that they can use the model.

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0




Age
Enter age
Monthly Spend
Monthly spend
Tenure Months
Months
Support Calls
Number of calls
Bank Transfer
Yes
Credit Card
Yes
Electronic Check
Yes
Predict
Model.pkl

Prediction Phase

Accept User Input

- Get input data from the user or application.

Preprocess the Input

- Apply the same preprocessing steps used during training.

Load the Saved Model

- Load the model.pkl model.

Make Prediction

- Use the trained model to predict the output.

Display the Result

- Return the prediction to the user.

For above we need to create the APT. the same will be created after model training.

Create an API

- Build an API (using Flask or FastAPI) so other applications can send data and use the trained model for predictions.

Data Collection

EDA

1) **Data Cleaning:**
2) **Data Analysis**
3) **Advance Pre-processing**
    - Handel Imbalance Data
    - Feature Selection
    - Dimension Reduction
4) **Encoding**
5) **Split the data Training and Testing**
    - **Training Data 80%,**
    - **Testing Data20%**
6) **5) Scaling**

Model training and Testing Pipeline

Prediction pipeline